Project : Project 5, Monitoring Kubernetes clusters with AI (Cloud Services and Systems, HES-SO Master)
Group : 13
Students : Nicolas Favre, Malik Fleury, Khalil El Ajeri
Git URL (if applicable) : https://gitlab.forge.hefr.ch/nicolas.favre1/k8s-observability (branch dev/nicolas)
Link to a video : [TO BE COMPLETED: video link]

Architecture

A monitoring assistant answers questions about a Kubernetes cluster in natural language: the engineer asks in Open WebUI, the assistant queries Mimir (metrics) and Loki (logs) with the help of the LLM, then answers and lists the tools it called. The figure shows the version deployed on AWS.

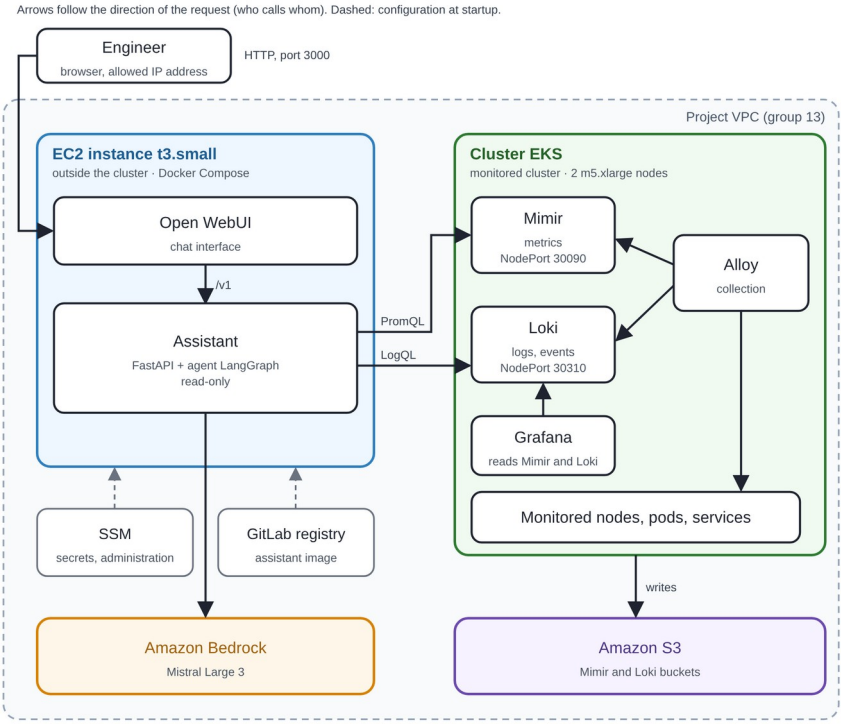


Figure: AWS deployment (us-east-1). The assistant runs on an EC2 instance, outside the EKS cluster it monitors.

Component	Role
Open WebUI	Chat interface where the engineer asks questions in French. It calls the assistant through an OpenAI-compatible API (/v1).
Assistant	FastAPI + LangGraph agent: turns the question into PromQL/LogQL calls (tools), summarises the result and lists the tools called with their arguments. Read-only; refuses off-topic requests.
LLM	Amazon Bedrock, Mistral Large 3 (public model). Local variant: Ollama, qwen3:8b (Kind only).
Alloy	Collects metrics and logs from nodes and pods (k8s-monitoring) and sends them to Mimir and Loki.
Mimir / Loki	Mimir stores metrics (PromQL), Loki stores logs (LogQL). Both write their data to S3.
Grafana	Dashboards and Explore on Mimir and Loki; used to replay the assistant's queries and check them.
Amazon S3	6 buckets for Mimir blocks and Loki chunks.

How a question travels

1. The engineer types a question in Open WebUI, for example "How is my cluster doing?".
2. Open WebUI passes it to our assistant, which looks like an ordinary OpenAI-style model to Open WebUI.
3. The assistant asks the LLM what to look up. The LLM picks tools (a PromQL query on Mimir, a LogQL search in Loki, or a pod description), and the assistant runs them.
4. The LLM writes the answer from the results. The assistant then adds the list of tools it called, with their exact arguments, so that anyone can replay the queries in Grafana.

Deployment

1. **Kubernetes cluster**: Amazon EKS (g13-p5-eks, 2 m5.xlarge nodes). Alloy, Mimir, Loki and Grafana are installed with Helm through OpenTofu (about 40 pods).
2. **Storage**: 6 Amazon S3 buckets, used by Mimir and Loki (writes observed on 28 Oct).
3. **Assistant and Open WebUI**: 2 Docker Compose containers on a single EC2 t3.small instance, outside the cluster. The assistant image comes from the HEFR GitLab registry; secrets are in AWS SSM Parameter Store; administration goes through SSM, with no SSH key.
4. **LLM**: Amazon Bedrock (Mistral Large 3), reached through the instance's IAM role, limited to this model (read in the code).
5. **Network**: no load balancer and no TLS. Open WebUI (HTTP, port 3000) accepts only the team's IP addresses; the assistant reads Mimir and Loki on ports 30090 and 30310 of one node, opened only to the host's security group (no authentication).
6. **Automation and cost**: OpenTofu and scripts (make aws-up: 19 min 12 s on 27 Oct; make aws-down; make aws-audit). Nodes and control plane: about $0.48/h (estimated from list prices, never checked on an invoice).

Tasks

Tasks	Outcome	Students involved
Observability stack and assistant, local (Kind)	Mimir, Loki, Alloy, Grafana, Open WebUI and assistant; smoke test 6 out of 6	Nicolas
Deployment on SWITCHengines (20 Oct)	2 VMs and k3s with OpenTofu, S3, stack deployed in 12 min 35 s, smoke test 6 out of 6, then deleted (part 2 report)	Khalil
Local hybrid mode with the LLM on Bedrock (25 Oct)	LLM_PROVIDER=bedrock variable, secret script, adapted smoke test	Malik
Deployment on AWS (27 Oct), after the professor's feedback of 22 Oct	EKS, EC2 host, Bedrock, S3; deployment, destroy and audit scripts	Nicolas
Assistant, tests and improvement plan	Assistant 0.5.x; 15 test scenarios on AWS (27-28 Oct); plan of 15 fixes	Nicolas
Documents and demonstration	Architecture v1 to v4 (with requirements), guide, demo script, video	Nicolas

Results of the tests

15 scenarios on AWS, 27-28 October, a handful of trials each, judged by hand against kubectl or direct queries. These numbers show what happened, not general reliability.

Scenario	What we saw
Cluster health	Node and pod counts correct in 5 trials out of 6. The first trial said "0 pods" because the model invented a metric; fixed afterwards.
Incident on a pod	Right pod, restart count and error message. Some durations were misread, and the logs of a pod that crashes in a loop were only partly collected.
Root cause	Right cause in 7 valid trials out of 7. Facts and guesses were cleanly separated in 3 trials, only partly in 4.
Node and service	Node state right in 4 trials out of 6, usually with one wrong or unchecked claim. Asked about a service, it described a pod instead.
Refusals and safety	Off-topic questions: 20 refused out of 20. Requests to delete or restart something: 3 out of 3 refused. Requests for secrets: 2 out of 2 refused. An instruction hidden in logs was not obeyed (2 out of 2). A joke mixed with a real question was answered without an explicit refusal of the joke.
Failures	With Mimir's query-frontend deleted, the assistant said Mimir was unreachable and invented no number; the pod was back in about 55 s. Restarting the assistant took about 2 s and the open conversation kept working. With a wrong model ID, the raw AWS error was shown.

Why these choices

Choice	Why	Alternatives set aside
AWS (EKS, S3, EC2)	On 22 Oct the professor told us the group has an AWS space with credits. EKS is AWS's managed Kubernetes and S3 its native object store.	Switch Cloud (managed Kubernetes), dropped. Our SWITCHengines test only gave VMs, so k3s had to be set up on them.
Alloy, Mimir, Loki, Grafana; LangChain, FastAPI	Required by the project statement, so we studied no alternative.	None studied.
OpenTofu and Helm, one module for local and cloud	The professor asks for automation. One module stops the local and cloud setups from drifting apart: only the cluster and S3 change. A full AWS deployment takes 19 min 12 s.	None documented.
Bedrock (Mistral Large 3) on AWS, Ollama (qwen3:8b) locally	The professor asked for a public model: local results were weak and resources costly. Mistral Large 3 costs $0.50 in and $1.50 out per million tokens (list price, 22 Oct). Locally nothing leaves the machine, but a CPU gives only 9 to 12 tokens per second.	A self-hosted LLM in the cloud, ruled out by the professor.
One LangGraph agent behind one endpoint	We keep control of the queries and can test without the LLM. After seeing invented queries and wrong units, we moved judgment from the model to Python code.	None documented.
Assistant on an EC2 t3.small, outside the cluster	The professor asks to keep what measures apart from what reads the measures. An instance is created and deleted by script and billed by the hour. Memory is tight: 489 MiB free out of 1909 at rest.	ECS Fargate, Lambda, the group's Mac.
Access through SSM, secrets in Parameter Store	No port 22 to open and no private key to look after. Secrets are kept out of the instance's startup data, where its description would show them.	An SSH key pair.
Read-only assistant that shows its tools	Safer: it only sends GET requests to Mimir and Loki, with no kubectl and no write right. An answer you can check is worth more than a fluent one you cannot. We replayed one query from the footer in Grafana: same pod, 262.3 MiB against 264.4 MiB, 14 minutes apart. It is not a guarantee: some answers still held wrong claims.	None documented.
2 m5.xlarge nodes for EKS	Upper bound taken from the Kind install (7 to 8 GiB at most). m5 rather than t3 for a predictable cost (about $0.05/h more). Real consumption was not measured.	t3 nodes.

Open problems and next steps

1. **Open problems**: no tool for alerts (Mimir's ruler and Alertmanager are empty), it can read an empty result as "nothing there", and Mimir and Loki are not monitored themselves. Our improvement plan lists 15 fixes; none is done yet.
2. **Cost and end of project**: the cluster is billed while it exists. We plan to destroy it on 30 October with make aws-down, check with make aws-audit, and let the video serve as the record.

Remarks. Nicolas did not test Malik's and Khalil's implementations: their contribution is described from their Git commits and Khalil's report. All three students used AI tools.